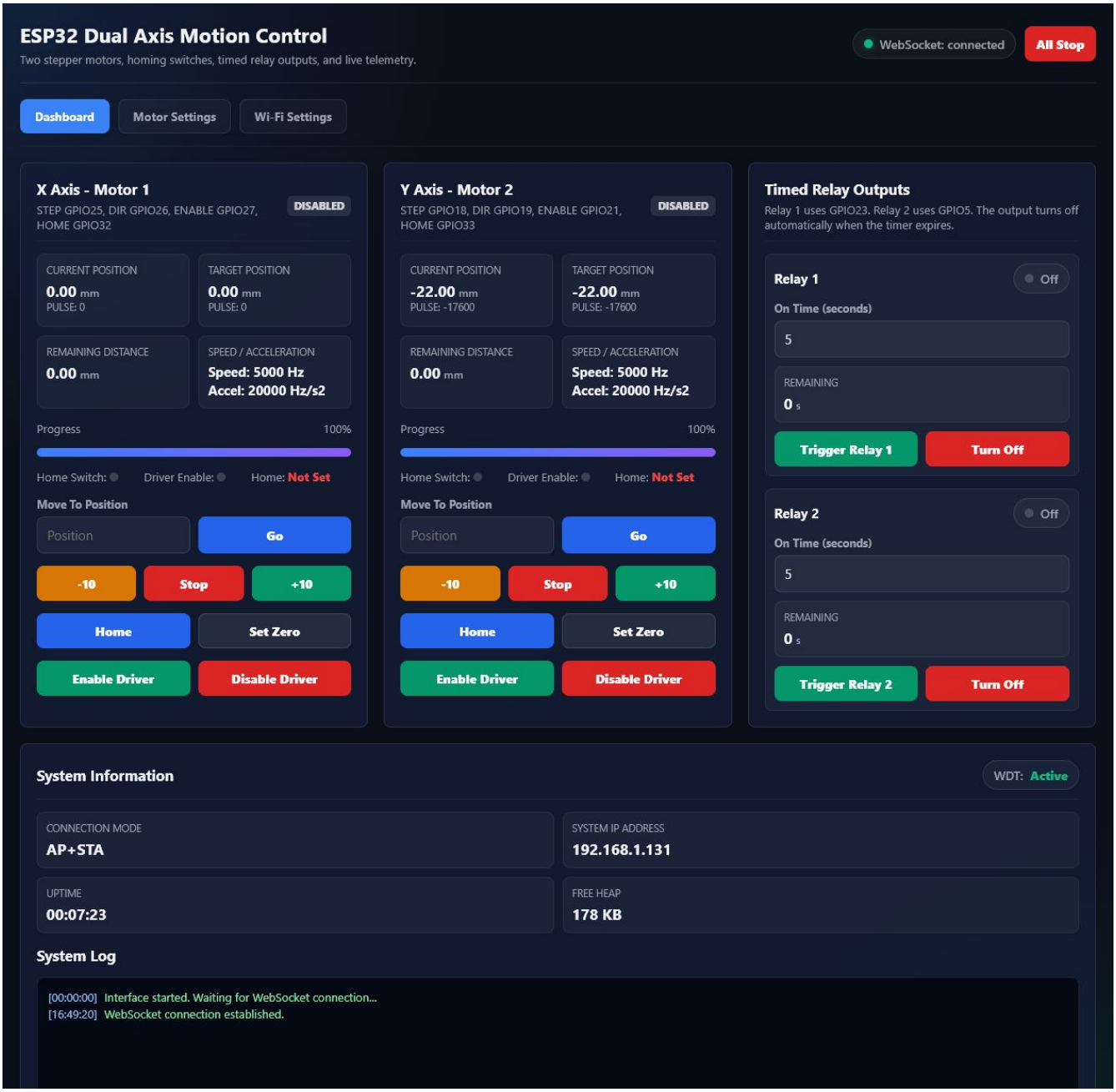
ESP32 WiFi Dual Axis Motion Control

Setup and operation guide for a two-axis DRV8825 stepper controller with an embedded wireless web dashboard, AP mode access, optional STA mode network access, homing inputs, and timed relay outputs.



Live dashboard screenshot captured from the ESP32 web interface.

System Overview

This firmware controls two independent stepper axes from an ESP32. Each axis uses a DRV8825 driver with STEP, DIR, ENABLE, and microstep selection signals. The web dashboard provides movement commands, live position telemetry, homing, motor setup, Wi-Fi configuration, system status, and two timed relay outputs.

Wireless control is available in both ESP32 AP mode and STA mode. AP mode lets the ESP32 create its own network for direct access. STA mode lets the ESP32 join an existing Wi-Fi router after credentials are saved, so the same dashboard can be opened from the router-assigned ESP32 IP address.

Required Hardware

- ESP32 WROOM or compatible ESP32 DevKit board.
- Two DRV8825 stepper driver modules.
- Two stepper motors such as NEMA14, sized for the selected mechanics and supply.
- Two normally-open or normally-closed home switches wired as active-low inputs.
- Two relay module inputs or transistor driver stages for the timed outputs.
- A motor power supply suitable for the DRV8825 VMOT input and motor current.

Pin Connections

Function	ESP32 GPIO	Connection
Motor 1 STEP	GPIO25	DRV8825 Motor 1 STEP
Motor 1 DIR	GPIO26	DRV8825 Motor 1 DIR
Motor 1 ENABLE	GPIO27	DRV8825 Motor 1 ENABLE
Motor 1 HOME	GPIO32	Home switch signal, other terminal to GND
Motor 1 M0 / M1 / M2	GPIO14 / GPIO13 / GPIO4	DRV8825 M0 / M1 / M2
Motor 2 STEP	GPIO18	DRV8825 Motor 2 STEP
Motor 2 DIR	GPIO19	DRV8825 Motor 2 DIR
Motor 2 ENABLE	GPIO21	DRV8825 Motor 2 ENABLE
Motor 2 HOME	GPIO33	Home switch signal, other terminal to GND
Motor 2 M0 / M1 / M2	GPIO16 / GPIO17 / GPIO22	DRV8825 M0 / M1 / M2
Relay 1 signal	GPIO23	Relay module IN1 or transistor driver input
Relay 2 signal	GPIO5	Relay module IN2 or transistor driver input

Power And Ground Notes

- Connect ESP32 GND, both DRV8825 logic grounds, relay module GND, home switch GND, and motor supply GND together.
- Connect the motor supply positive output to each DRV8825 VMOT input, not to the ESP32 3.3 V pin.
- Set the DRV8825 current limit before connecting the motors to the mechanics.
- Use a relay module or transistor driver. Do not drive relay coils directly from ESP32 GPIO pins.

Wireless Access

On first boot or when no router credentials are configured, use AP mode. Connect your computer or mobile device to SSID ESP32-MOTOR with password 12345678, then open http://192.168.4.1 in a browser.

To use STA mode, open the Wi-Fi Settings tab, enter the router SSID and password, save the settings, and reboot the ESP32. After reboot, the board continues to provide AP access and also attempts to join the configured network. When STA is connected, use the IP address shown in the dashboard system panel or serial monitor.

Dashboard Operation

- Use the Dashboard tab for absolute moves, jog moves, homing, driver enable control, and relay triggering.
- Use the Motor Settings tab to configure speed, acceleration, travel limits, step angle, microstepping, mechanics, display units, and auto-disable timing.
- Use the Wi-Fi Settings tab to save STA credentials for router-based wireless access.
- Use the Relay Output controls by entering an on-time in seconds and pressing the trigger button. The relay turns off automatically when the timer reaches zero.

Build And Upload

- Install the ESP32 board package in Arduino IDE or Arduino CLI.
- Install FastAccelStepper, ESPAsyncWebServer, AsyncTCP, and ArduinoJson.
- Open main/main.ino and confirm the pin assignments in main/Config.h.
- Compile for an ESP32 Dev Module compatible board profile and upload to the ESP32.
- After upload, connect through AP mode or STA mode and test each axis at low speed first.

Safety

- Use fusing and suitable wiring for the motor and relay load currents.
- Add hardware emergency stop and limit protection for any machine that can cause damage or injury.
- Keep relay-controlled AC or high-current loads isolated inside a proper enclosure.
- Verify all motion directions and home switch behavior before normal operation.